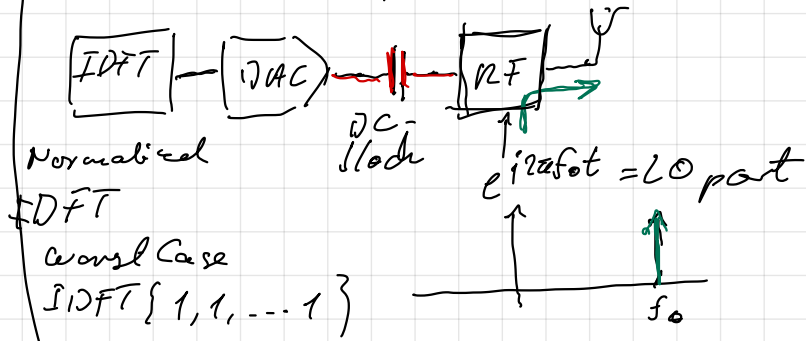
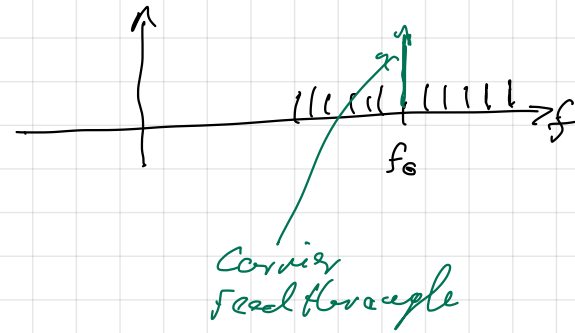
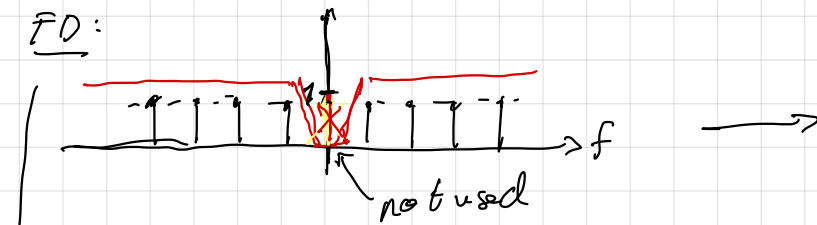


PAPR - problem : Peak-to-Average Power Ratio

FD:

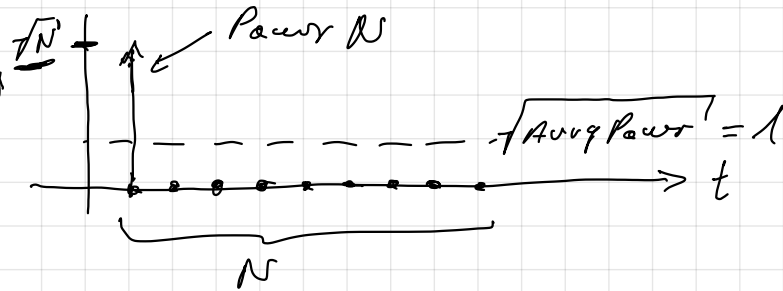


Normalised

IDFT

const Case

IDFT $\{1, 1, \dots, 1\}$



$$\text{PAPR} = N$$

OFDM Channel Estimation (wide band)

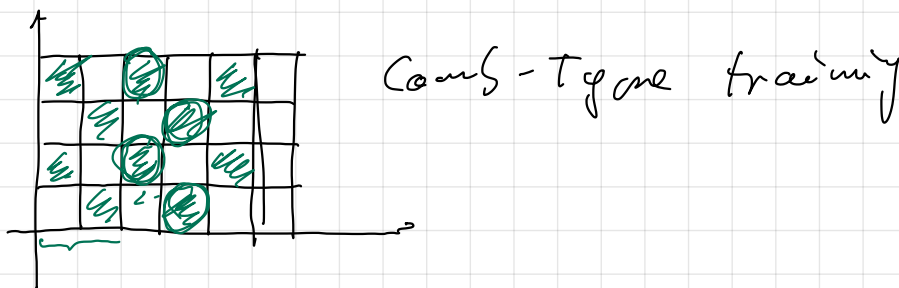
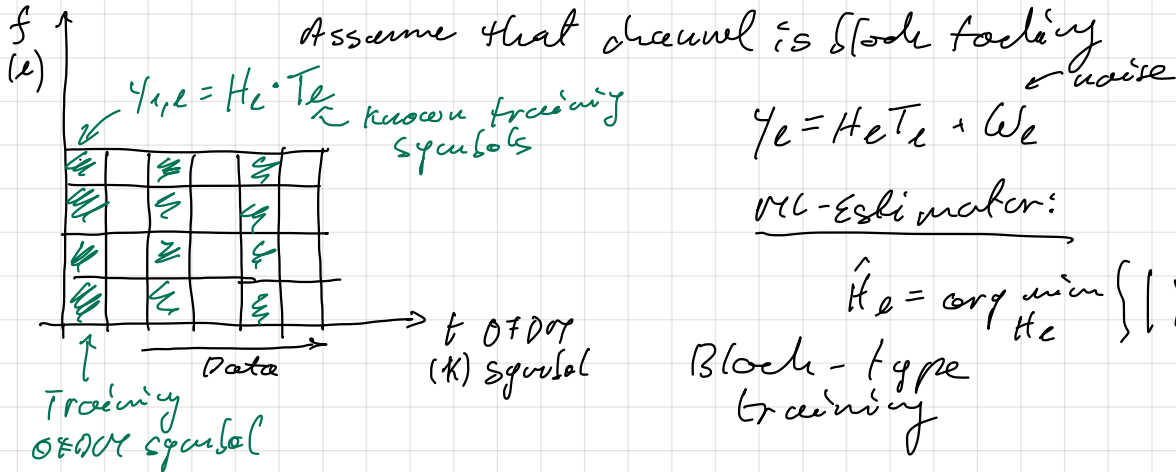
Channel is NOT scalar

Per-tone channel estimate:

$$y_e = H_e a_e$$

↑
need to know H_e

Training based channel estimation



Channel estimation error:

$$\mathbb{E} \left\{ |\hat{H}_e - H_e|^2 \right\} = \mathbb{E} \left\{ \left| \frac{y_e}{T_e} - H_e \right|^2 \right\}$$

$$= \mathbb{E} \left\{ \left| H_e + \frac{w_e}{T_e} - H_e \right|^2 \right\} = \mathbb{E} \left\{ \left| \frac{w_e}{T_e} \right|^2 \right\} = \frac{1}{|T_e|^2} \mathbb{E} \{ |w_e|^2 \}$$

Noise variance

How to choose T_e ?

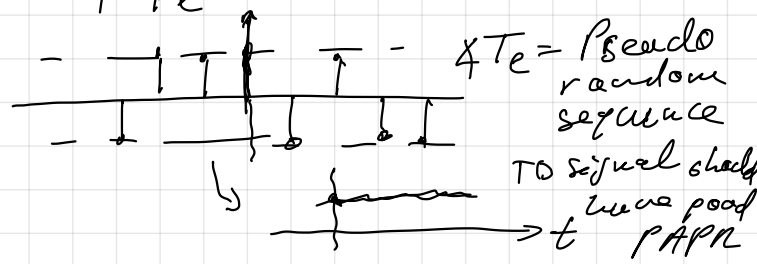
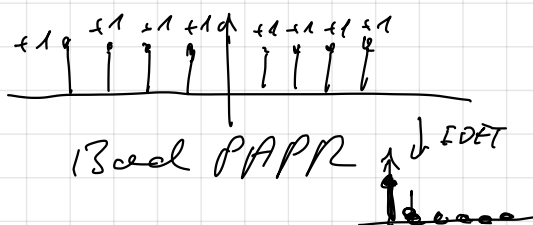
Power constraint: $\frac{1}{N} \sum |T_e|^2 = 1$

Min est. error $\frac{1}{N} \sum \left| \frac{\sigma_w}{T_e} \right|^2 \rightarrow \min$

$\Rightarrow$ Optimum if all $|T_e|$ equal

$\rightarrow$ What is left to choose is the

phase of T_e



Performance impact of the channel est. error

Set $\Delta H_e = H_e - \hat{H}_e$
↑
ch. est. error

Input-output-relationship: $y_e = H_e a_e + w_e = (\hat{H}_e + \Delta H_e) a_e + w_e$
during data phase

$$\hat{H}_e + \Delta H_e \quad = \hat{H}_e a_e + \underbrace{\Delta H_e a_e + w_e}_{\text{Noise} \rightarrow \bar{w}_e}$$

Noise: $\bar{w}_e = \Delta H_e a_e + w_e$ ← assume ind. of a_e

Noise variance of $\bar{w}_e$: $E\{|\bar{w}_e|^2\} = \underbrace{E\{|\Delta H_e|^2\}}_{\sigma_w^2} \underbrace{E\{|a_e|^2\}}_1 + \underbrace{E\{|w_e|^2\}}_{\sigma_w^2}$
 $= 2\sigma_w^2$

compare to perfect ch. est.: Noise doubles
 $\sim \text{SNR degrades} \sim 3 \text{ dB}$

Least squares channel estimate

- start with OFDM system with N tones
- All tones are used for training
- Channel impulse response $h[\tau] = 0$, $\tau \gg L$ (i.e. L non-zero taps or unknowns) $L \ll N$
to be efficient

Per-tone estimate: • independent ch. estimate for all N tones from N observations

- Channel has only $L \ll N$ non-zero-taps (degrees of freedom)
 $\Rightarrow H_e$ must be correlated

Idea: estimate only $h[0] \dots h[L-1]$ (L unknowns)

1) $\underline{Y} = \underline{I} \underline{H}$ $\underline{I} = \text{diag}(T_e)$

$$\underline{I} = \begin{bmatrix} T_0 & \tau_1 & 0 \\ & \ddots & \\ 0 & & T_{N-1} \end{bmatrix} \quad \underline{H} = \begin{bmatrix} H_0 \\ \vdots \\ H_{N-1} \end{bmatrix}$$

2) write $\underline{H}$ as function of TD impulse response

$\underline{h} = [h_0 \dots h_{L-1}]^T$ using the Fourier Matrix

$$\underline{H} = \underline{F}_{N \times N} \cdot \begin{bmatrix} h_0 \\ \vdots \\ h_{L-1} \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \underline{F}_{N \times L} \begin{bmatrix} h_0 \\ \vdots \\ h_{L-1} \end{bmatrix} \quad \underline{F}_{N \times L} = \text{first } L \text{ columns of } \underline{F}_{N \times N}$$

$$3) \underline{Y} = \underline{T} \underline{H} = \underline{T} \underline{F}_{N \times L} \underline{h}$$

N : equations
 $L \ll N$ unknowns

solve for $\underline{h}$ in a least squares sense

$$\hat{\underline{h}} = (\underline{T} \underline{F}_{N \times L})^+ \underline{Y} = (\underline{F}_{N \times L}^H \underline{T}^H \underline{T} \underline{F}_{N \times L})^{-1} \underline{F}_{N \times L}^H \underline{T}^H \underline{Y}$$

$$\hookrightarrow \hat{H}_2 = \text{IDFT} \{ \hat{\underline{h}} \}$$

$$\underline{Y}' = \begin{bmatrix} y_1 \\ y_2 \\ y_4 \\ \vdots \\ y_{N-1} \end{bmatrix} : \text{only every 2nd tone} \quad L \ll \frac{N}{2}$$

$$\underline{F}_{N/2 \times L} = \begin{bmatrix} \checkmark \\ \times \\ \checkmark \\ \times \end{bmatrix} \quad \leftarrow \begin{array}{l} \text{omitted} \\ \text{every} \\ \text{2nd row} \end{array}$$